CATEGORICAL DATA ANALYSIS
PSYC 6136

YORK UNIVERSITY

Project 2

Submitted To:

Professor Michael L. Friendly,
Department of Psychology,
Room 226, Behavioural Sciences Building

Submitted By:

Gavin M. Klorfine,
Department of Psychology
Room 059, Behavioural Sciences Building

March 31st, 2026

Contents

Question 1: The Dayton Survey Data (`DaytonSurvey`)

The `DaytonSurvey` dataset from the vcdExtra package in R contains a $2 \times 2 \times 2 \times 2 \times 2$ contingency table containing the result of a 1992 survey that classified $N = 2276$ high school seniors by the following variables (Agresti, 2002; Friendly, 2026; R Core Team, 2025):

1. `alcohol` with levels `Yes` and `No` corresponding to whether the student had ever used alcohol.
2. `cigarette` with levels `Yes` and `No` corresponding to whether the student had ever used a cigarette.
3. `marijuana` with levels `Yes` and `No` corresponding to whether the student had ever used marijuana.
4. `sex` with levels `female` and `male` corresponding to the student's sex.
5. `race` with levels `white` and `other` corresponding to the student's race.

The goals of the subsequent analyses were to understand the associations among drug use variables `alcohol`, `cigarette`, and `marijuana`, and then to determine how these associations differ according to demographic variables `sex` and `race`.

Method

First, for a cursory glance of the structure of association between variables and their levels, a multiple correspondence analysis (MCA) was conducted (including all variables/levels). This was done in R (v. 4.5.3) using the package ca (v. 0.71.1; Greenacre & Nenadic, 2018; R Core Team, 2025).

Second, focusing on just the drug variables (i.e., `alcohol`, `cigarette`, and `marijuana`), mosaic and fourfold displays were constructed to illustrate patterns of association. This was done in R (v. 4.5.3) using the vcd (v. 1.4-13) and vcdExtra (v. 0.71.1) packages (Meyer, 2024; R Core Team, 2025).

Third, a loglinear model was fit to explain the structure of associations among the drug variables. This was done in R (v. 4.5.3) using the MASS package (v. 7.3-65; Ripley & Venables, 2025; R Core Team, 2025). The resulting model was depicted via a mosaic display to provide evidence regarding model fit.

Fourth, this analysis was extended by fitting a loglinear model that included both drug and demographic variables. This was done in R (v. 4.5.3) using the MASS package (v. 7.3-65; Ripley & Venables, 2025; R Core Team, 2025). The resulting model was depicted via a mosaic display to provide evidence regarding model fit.

Last, of included drug types in the `DaytonSurvey` dataset, `alcohol` and `marijuana` were the only intoxicants (i.e., `cigarette`s are not intoxicating); moreover, the association between `alcohol` and `marijuana` was thought to have particular importance. To analyze this association in relation to all of the other variables (i.e., `cigarette`, `sex`, and `race`), a main-effects analysis of variance (ANOVA) model was fitted using log odds ratios (LORs) and weights inversely proportional to the square of estimated sampling variances. Interaction effects were not included to avoid near-saturation of the model (a full model would have one less parameter than the number of strata). Additionally, LORs were plotted in a dot-and-whisker-plot, with whiskers corresponding to 95% confidence intervals (CIs).

Results

Multiple Correspondence Analysis

To explore the nature of the associations between the levels of variables in the `DaytonSurvey` dataset, an MCA was conducted. The first dimension accounted for 57.44% of the total inertia and the second dimension accounted for 16.85% of the total inertia. With both dimensions combined having accounted for a strong 74.29% of the total inertia, a 2D depiction of the MCA was meaningful. Thus, the MCA was plotted using the Burt matrix, with the result displayed in Figure 1.

Because both all `"Yes"` responses had negative values on dimension one and all `"No"` responses had positive values on dimension one, this dimension may be interpreted as that of "drug use to drug abstinence." In contrast, because `white` and `female` lay on the positive side of dimension two, with `male` and `other` laying on the negative side of dimension two, this dimension may be thought of as a combination of "femininity to masculinity," and "white to racialized."

Demographic variables `sex` and `race` had all levels at near-zero values on dimension one, apart from `race = "other"`. Moreover, with `"other"` lying on the negative side of dimension one, racialized students had similar profiles to responses of abstinence. Demographic levels of `"female"`, `"white"`, and `"male"` had response profiles more similar to those of drug use than those of drug abstinence. Furthermore, abstinent responses had similar profiles and responses of drug use had similar profiles.

Mosaic and Fourfold Displays

To illustrate patterns of association between the drug variables in the `DaytonSurvey` dataset, pairwise mosaic and fourfold displays were constructed (see Figures 2 and 3).

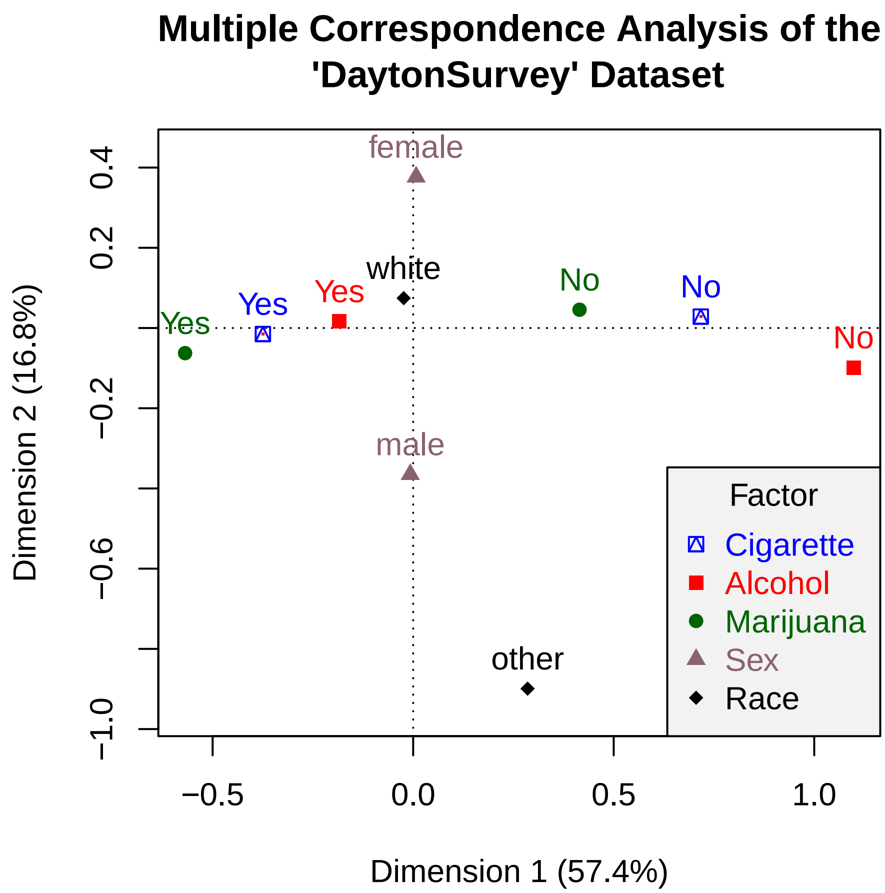

Figure 1. A biplot of the largest two dimensions of a multiple correspondence analysis conducted on the `DaytonSurvey` dataset. The proximity of points to one another reflects a similarity in response profiles. Adapted from "Discrete Data Analysis with R," by M. Friendly & D. Meyer, 2016, CRC Press, p. 245.

The pairwise mosaic displays of Figure 2 universally indicated strong associations with patterns of agreement. By this, it is meant that in all pairwise mosaics there exists intense shading (reflecting large residuals), with responses of `"Yes"` to having used a given drug coinciding with other responses of `"Yes"`, and responses of `"No"` coinciding with other responses of `"No"`.[1] For example, the marginal association between `cigarette` and `marijuana` usage (collapsing over all other variables) was strong, as indicated by the intense shading of the corresponding mosaic in Figure 2. Those who said `"Yes"` to having used `marijuana` also said `"Yes"` to having used `cigarette`s at a greater rate than would be expected under a model of independence (where `cigarette` and

[1] The pairwise association between `alcohol` and `marijuana` was strong, but weaker than the other pairwise associations (reflected by the less intense shading of negative residuals).

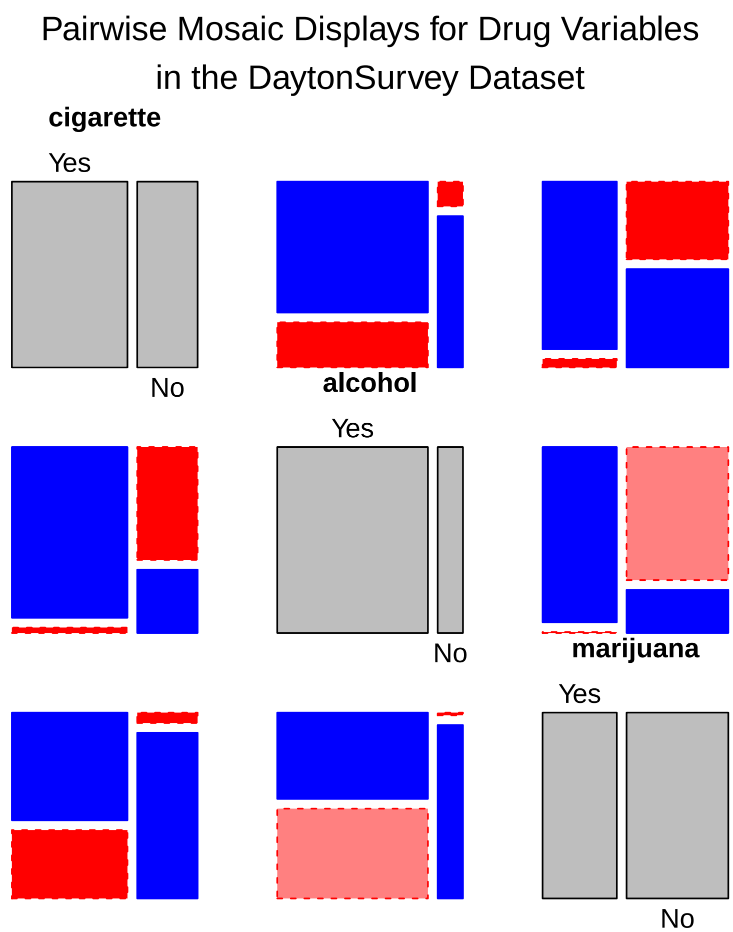

Figure 2. Pairwise Mosaic Displays for drug variables in the `DaytonSurvey` Dataset. Diagonal panels show marginal frequencies. Off-diagonal panels show the bivariate marginal relationship between the row and column variables (Friendly & Meyer, 2016). Cell Pearson residuals between an absolute value of two and four are lightly shaded, with darker shading corresponding to residuals greater than an absolute value of four. Smaller residuals are left un-shaded (i.e., those with absolute value less than two).

`marijuana` are not associated, given all other variables). Likewise, those who said `"No"` to having used `cigarettes` also said `"No"` to having used `marijuana` at a greater rate than would be expected under a model of independence.

Similarly, the fourfold displays of Figure 3 universally indicate strong associations between drug variables (i.e., odds ratios (ORs) not equal to 1), as illustrated through non-overlapping 95% CI bands.[2] First, among students that have used `alcohol`, those that additionally have used `cigarettes` were

[2]Displays with wide 95% CI bands should be interpreted with caution due to variable estimates produced by the low number of observations in certain cells.

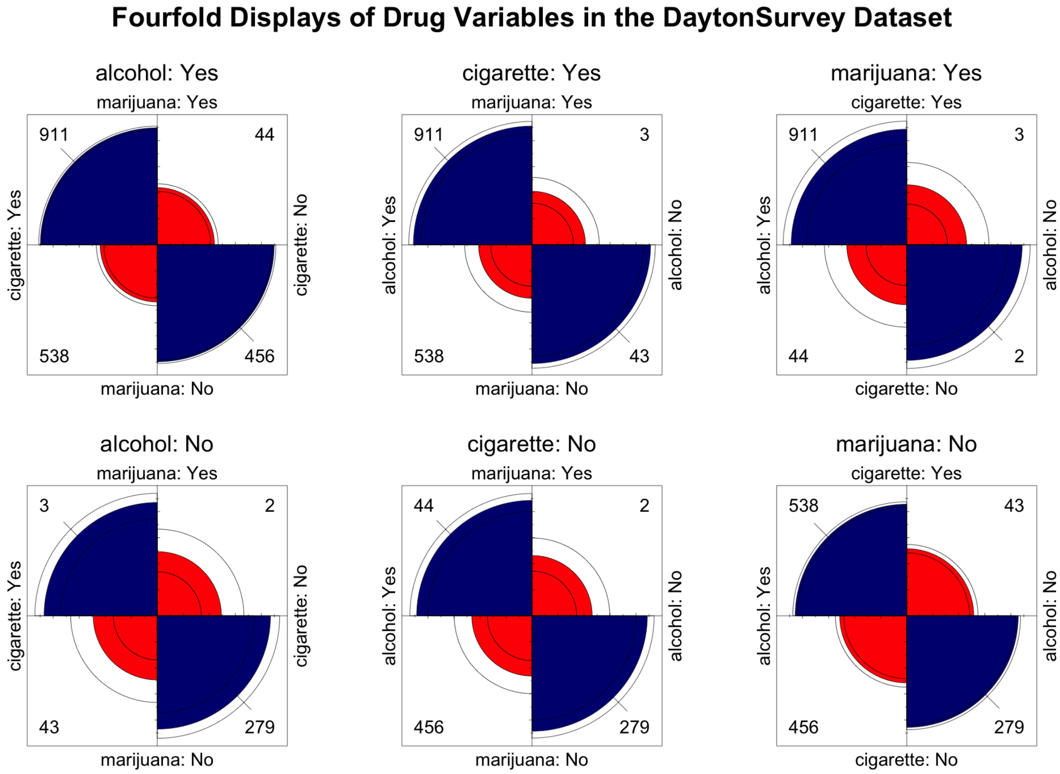

Figure 3. Fourfold displays of drug variables in the `DaytonSurvey` dataset. A quadrant's area is the standardized frequency in the corresponding cell. Intense shading and non-overlapping CI bands indicate a significant association (i.e., an odds ratio not equal to one).

17.55 times more likely to also have used `marijuana`. Second, among students that have not used `alcohol`, those that have used `cigarettes` were 9.73 times more likely to also have used `marijuana`. Third, among students that have used `cigarettes`, those that have additionally used `alcohol` were 24.27 times more likely to also have used `marijuana`.

Fourth, among students that have not used `cigarettes`, those that have used `alcohol` were 13.46 times more likely to also have used `marijuana`. Fifth, among students that have used `marijuana`, those that have additionally used `alcohol` were 13.8 times more likely to also have used `cigarettes`. Last, among students that have not used `marijuana`, those that have used `alcohol` were 7.66 times more likely to also have used `cigarettes`.

Overall, both Figures 2 and 3 point toward a pattern of poly-use; moreover, the use of one type of drug is associated with the additional use of one or both of the other included drug types.

Loglinear Models

To fit a loglinear model to the drug variables in the `DaytonSurvey` dataset, the model of mutual independence (i.e., [`alcohol`][`cigarette`][`marijuana`]) was taken as the starting point, with additional associations added until a reasonable fit was obtained. The starting model resulted in a bad fit, $G^2(4) = 1286.02$, $p < .001$, necessitating the addition of association terms. Next, the association of `alcohol` and `cigarette` was added, with the updated model resulting in a poor, yet significantly improved, fit ($G^2(3) = 843.83$, $p < .001$, $\Delta G^2(1) = 442.19$, $p < .001$). After this, a model of conditional independence was fit (i.e., [`alcohol`, `marijuana`][`cigarette`, `marijuana`]), with this model resulting in a poor, yet significantly improved, fit ($G^2(2) = 187.75$, $p < .001$, $\Delta G^2(1) = 656.07$, $p < .001$). Finally, the model of homogeneous association was fit (i.e., [`alcohol`, `cigarette`][`alcohol`, `marijuana`][`cigarette`, `marijuana`]), resulting in both a well-fitting model and a significant improvement over the previous model, $G^2(1) = 0.37$, $p = .541$, $\Delta G^2(1) = 187.38$, $p < .001$. The saturated model (i.e., the perfectly fitting model containing the four-way association) was not a significant improvement over the obtained model, $\Delta G^2(1) = 0.37$, $p = .541$.

Therefore, the fitted log-linear model was specified as [`alcohol`, `cigarette`][`alcohol`, `marijuana`][`cigarette`, `marijuana`]. Model fit is visualized in Figure 4 using a "clean" mosaic display (i.e., one with minimal to no shading as a result of plotting a reasonably well-fitting model; Friendly & Meyer, 2016).

Thus, associations were found in all pairwise combinations of drug variables, with these associations being constant across levels of the third drug (e.g., the association [`marijuana`, `cigarette`] would be constant across levels of `alcohol`). This interpretation is somewhat in conflict with that previously seen through the odds ratios. For example, with the association [`marijuana`, `cigarette`], the OR was 17.55 for responses of `"Yes"` to `alcohol`, whereas the OR was a substantially different 9.73 for responses of `"No"` to `alcohol`. This difference was likely an artefact of variable estimates produced by the low number of observations in certain cells. Hence, the given interpretations should be taken with caution.

To extend this log-linear model to include both drug *and* demographic variables (i.e., `sex` and `race`), the previous model plus an additional association term, [`sex`, `race`], was taken as the starting point, with additional associations added until a reasonable fit was obtained. The starting model resulted in a bad fit, $G^2(22) = 39.49$, $p = .012$, necessitating the addition of association terms. Next, the association of [`sex`, `race`, `alcohol`] was added, with the updated model resulting in both a marginal fit and one that was significantly improved over the previous model ($G^2(19) = 26.71$, $p = .112$, $\Delta G^2(3) = 12.79$, $p = .005$). Finally, adding an additional three-way association of [`sex`, `race`, `marijuana`] resulted in both a well-fitting model and a significant

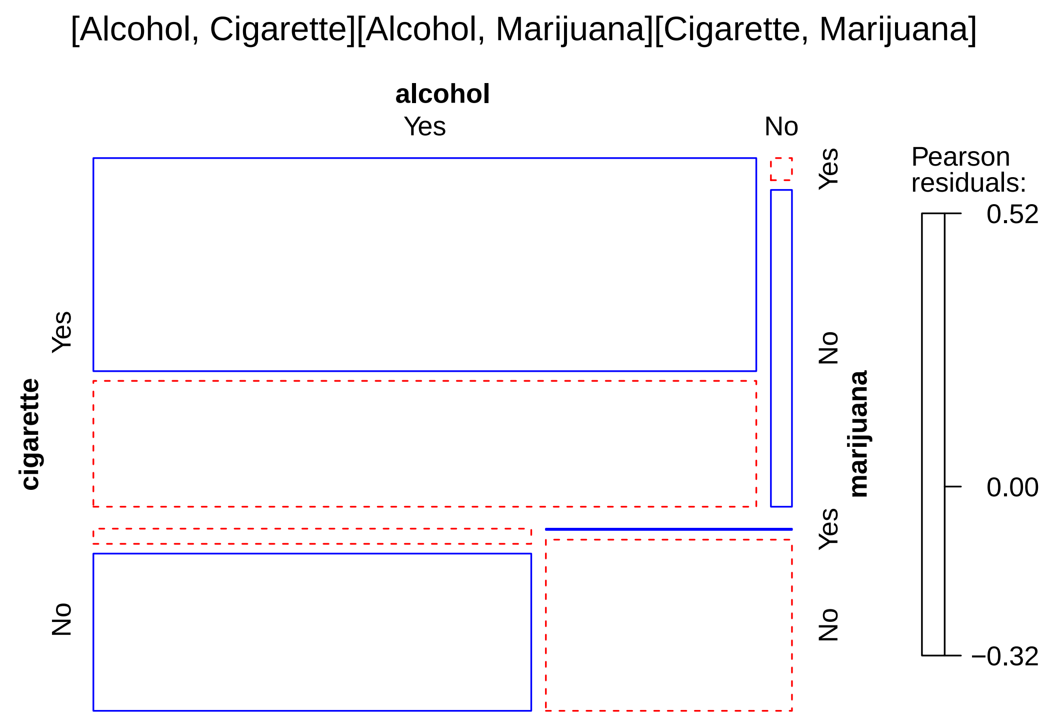

Figure 4. A "clean" mosaic display of the log-linear model specified as [`alcohol`, `cigarette`][`alcohol`, `marijuana`][`cigarette`, `marijuana`] (Friendly & Meyer, 2016).

---

improvement over the previous model, $G^2(16) = 14.66$, $p = .55$, $\Delta G^2(3) = 12.05$, $p = .007$. The saturated model (i.e., the perfectly fitting model containing the four-way association) was not a significant improvement over the obtained model, $\Delta G^2(16) = 14.66$, $p = .55$.

Therefore, the fitted log-linear model was specified as [`alcohol`, `cigarette`][`alcohol`, `marijuana`][`cigarette`, `marijuana`][`sex`, `race`, `alcohol`][`sex`, `race`, `marijuana`]. Model fit is visualized in Figure 5 using a "clean" mosaic display (i.e., one with minimal to no shading as a result of plotting a reasonably well-fitting model; Friendly & Meyer, 2016). Thus, this indicates that one's demographics (i.e., their `sex` and `race`) are associated with their decision to use each of `alcohol` and `marijuana`, but not with their decision to use `cigarettes`.

Analyses of Log Odds Ratios

To analyze the association between `alcohol` and `marijuana` in relation to all other variables (i.e., `cigarette`, `sex`, and `race`), LORs and their 95%

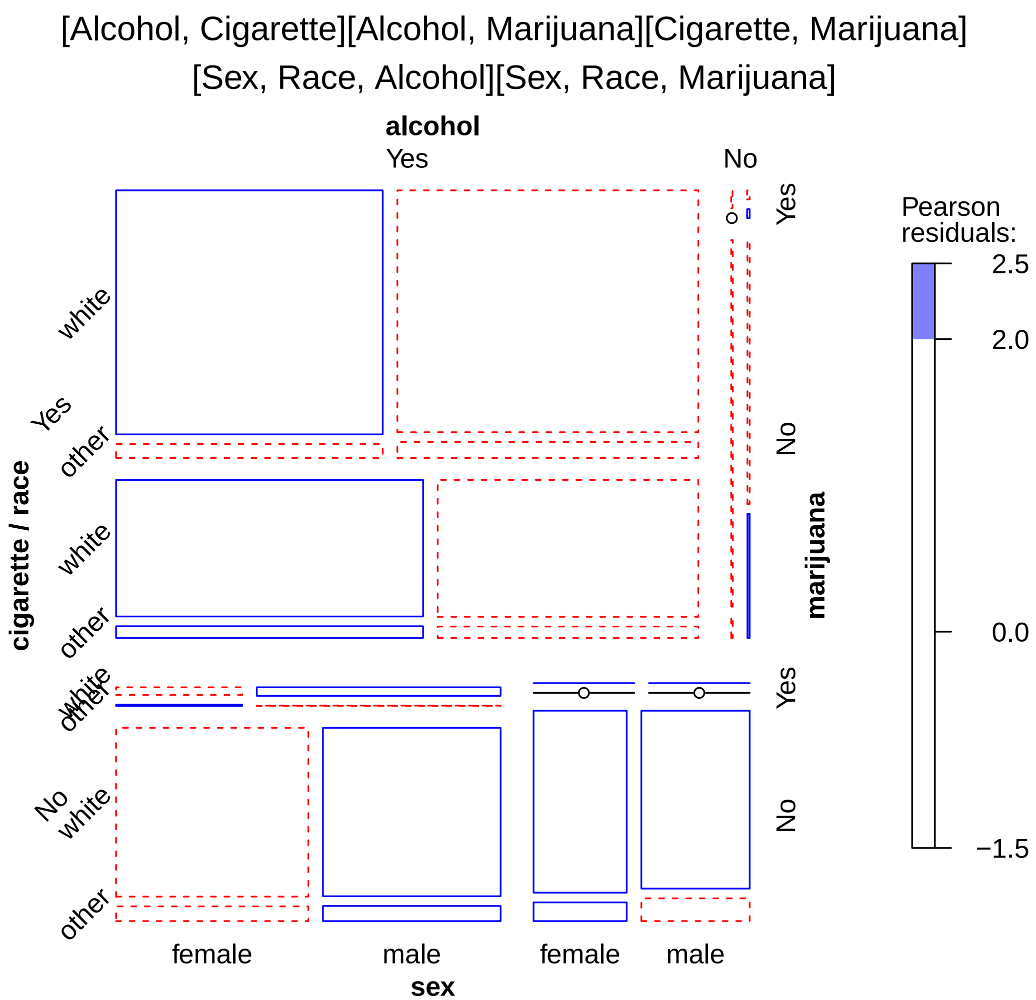

Figure 5. A "clean" mosaic display of the log-linear model specified as [`alcohol`, `cigarette`][`alcohol`, `marijuana`][`cigarette`, `marijuana`][`sex`, `race`, `alcohol`][`sex`, `race`, `marijuana`] (Friendly & Meyer, 2016).

CIs were calculated and then illustrated through the dot-and-whisker plot of Figure 6. In addition, a main-effects ANOVA was conducted on these LORs with weights inversely proportional to the square of each estimated sampling variance (see Method for the reasoning behind excluding interaction terms).

With white female students that have used `cigarette`s as the reference category, the model intercept (the mean of the reference category) was significantly different from zero, $\hat{\beta}_0 = 2.67$, 95% CI (2.1, 3.24), $t(4) = 13.04$, $p < .001$. With a sample LOR of 2.87, there was a strong association between `alcohol` and `marijuana` use in this category. The effect of `cigarette` use was significant, with abstinence resulting in a relatively strong -0.87 average decrease in the sample LOR ($\hat{\beta}_{\text{cigarette = "No"}} = -0.87,\ 95\%\ \text{CI}\ (-1.47,\ -0.26)$,

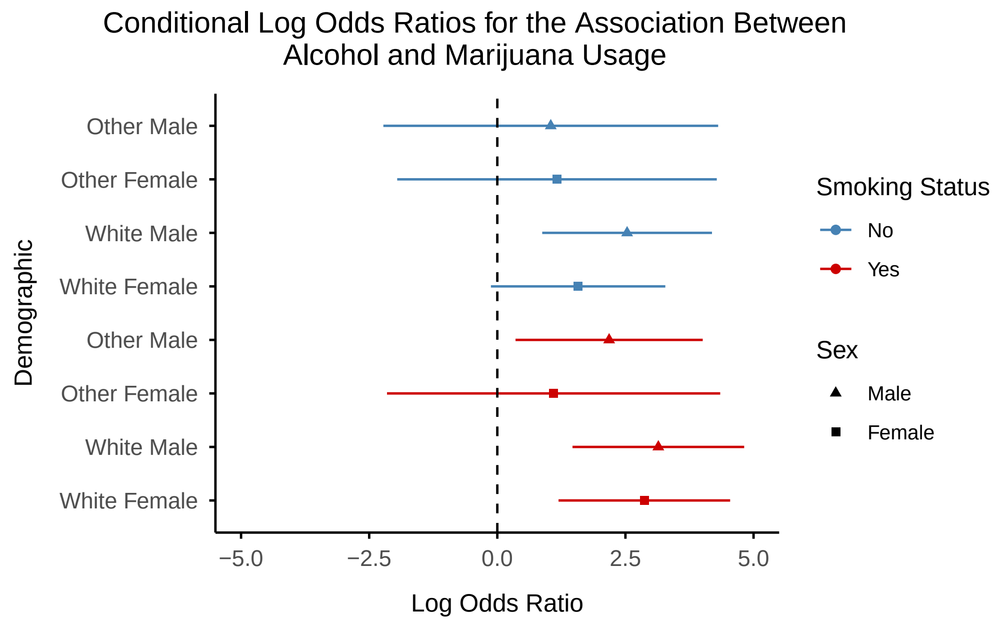

Figure 6. A dot-and-whisker plot of conditional LORs for the association between `alcohol` and `marijuana` across all combinations of other variables (i.e., `cigarette`, `sex`, and `race`). "Dots" represent calculated LORs, with whiskers representing the corresponding 95% CI.

$t(4) = -3.96$, $p = .017$). The effect of `sex` was (marginally) not significant, with males associated with a moderate 0.59 average increase in the sample LOR ($\hat{\beta}_{\text{sex = male}} = 0.59$, 95% CI $(-0.02,\ 1.2)$, $t(4) = 2.71$, $p = .054$). Last, the effect of `race` was significant, with non-whites (`"other"`) associated with a strong -1.12 average decrease in the sample LOR ($\hat{\beta}_{\text{race = "other"}} = -1.12$, 95% CI $(-1.79,\ -0.46)$, $t(4) = -4.68$, $p = .009$).

These main effects were apparent in Figure 6, with three of four strata that responded `"No"` to `cigarette` use having 95% CIs that overlap with LOR $= 0$ (i.e., no association between `alcohol` and `marijuana`). In contrast, ony one of four strata that responded `"Yes"` to `cigarette` use had 95% CIs that overlapped with LOR $= 0$. Additionally, of strata that had 95% CIs overlapping with LOR $= 0$, three of four were female, and three of four were non-white. Therefore, the association of `alcohol` and `marijuana` was stronger for `males`, those who have smoked a `cigarette`, and whites.

Summary

Results of these analyses point to a pattern of poly-substance use, where the use of either `alcohol`, `marijuana`, or `cigarette`, is associated with the use of one or both of the other drugs. Demographics of `sex` and `race` also were associated with the use of intoxicants (`alcohol` and `marijuana`), but not `cigarette`, with males and whites exhibiting larger strengths of the `alcohol` and `marijuana` association.

Future studies should gather a greater sample size so that per-cell ns are large enough to stabilize estimates, conduct a full ANOVA (i.e., one that includes interaction terms), and avoid conflicts of interpretation between analyses.

Question 2: The Housing Data (`housing`)

The `housing` dataset from the MASS package in R contains a $3 \times 3 \times 4 \times 2$ contingency table containing the result of a survey that classified $N = 1681$ Copenhagen residents by the following variables (Cox & Snell, 1984; Madsen, 1976; R Core Team, 2025; Ripley & Venables, 2025):

1. `Sat`: the **satisfaction** of residents with their current level of housing. Levels include `High`, `Medium`, and `Low`.

2. `Infl`: residents' perceived degree of **influence** on the management of the property. Levels include `High`, `Medium`, and `Low`.

3. `Type`: the kind of rented property. Levels include `Tower`, `Atrium`, `Apartment`, and `Terrace`.

4. `Cont`: the amount of **contact** that residents have with one another. Levels include `Low` and `High`.

Residents' `Sat`isfaction with their housing is plausibly associated with their perceived degree of `Infl`uence, the amount of `Cont`act they have with other residents, and the `Type` of property they are renting. Thus, the following analyses aim to explain how `Sat` varies with `Infl`, `Cont`, and `Type`.

Method

For a cursory glance of the structure of association between variables and their levels, an MCA was conducted (including all variables/levels). This was done in R (v. 4.5.3) using the package `ca` (v. 0.71.1; Greenacre & Nenadic, 2018; R Core Team, 2025).

Next, a proportional odds (PO) model for ordinal logistic regression was fitted to the data. This was done in R (v. 4.5.3) through the MASS package (v. 7.3-65) by first fitting a main-effects model and then performing a stepwise model selection procedure by Akaike's An Information Criterion (AIC) to obtain significant interaction terms (procedure performed in the forward direction; Ripley & Venables, 2025; R Core Team, 2025). The PO assumption was tested by comparing the created model to one that allowed for non-parallel slopes. This was done through both a likelihood ratio test in the VGAM (v. 1.1-14) package, as well as a novel graphical method (R Core Team, 2025; Yee, 2025). Next, obtained interaction terms were probed in R (v. 4.5.3) using the emmeans package (v. 2.0.1), with *p*-values adjusted via the Holm method and CIs being left unadjusted (Lenth & Piaskowski, 2025; R Core Team, 2025). [3] Lastly, model results were illustrated through both plots of predicted probabilities and effect plots. The effect plots were created in R (v. 4.5.3) using the effects package (v. 4.2-5; Fox et al., 2026; R Core Team, 2025).

Results

Multiple Correspondence Analysis

To explore the nature of the associations between the levels of variables in the housing dataset, an MCA was conducted. The first dimension accounted for 22.27% of the total inertia and the second dimension accounted for 15.52% of the total inertia. With both dimensions combined having accounted for a moderate 37.79% of the total inertia, a 2D depiction of the MCA was meaningful. Thus, the MCA was plotted using the Burt matrix, with the result displayed in Figure 7.

The first dimension captured the ordering of both Sat and Infl (i.e., levels of Low were negative, levels of Medium were near zero, and levels of High were positive). This suggests that dimension one may be interpreted as a combination of housing satisfaction and degree of influence. On the other hand, dimension two appeared to be associated with residents' degree of Contact with each other.

The variable of rental property Type appeared to be associated with both dimensions: on dimension one the level Terrace had a negative value, levels Apartment and Atrium had near-zero (i.e. "Medium") values, and Tower had a positive value. On dimension two, the level Atrium had a large, negative value, levels Terrace and Apartment had near-zero values, and the level Tower had a positive value.

High Satisfaction was most similar in profile to High perceived degree of Influence, and both the Tower and Apartment property Types. High Satisfaction was in-between Contact levels, suggesting that moderate levels of contact with other residents affords satisfaction, with "too much" (i.e., High) contact being similar in profile to Medium satisfaction, and low contact being similar in profile to Low satisfaction.

[3] Thus, CIs should be interpreted descriptively (and not inferentially).

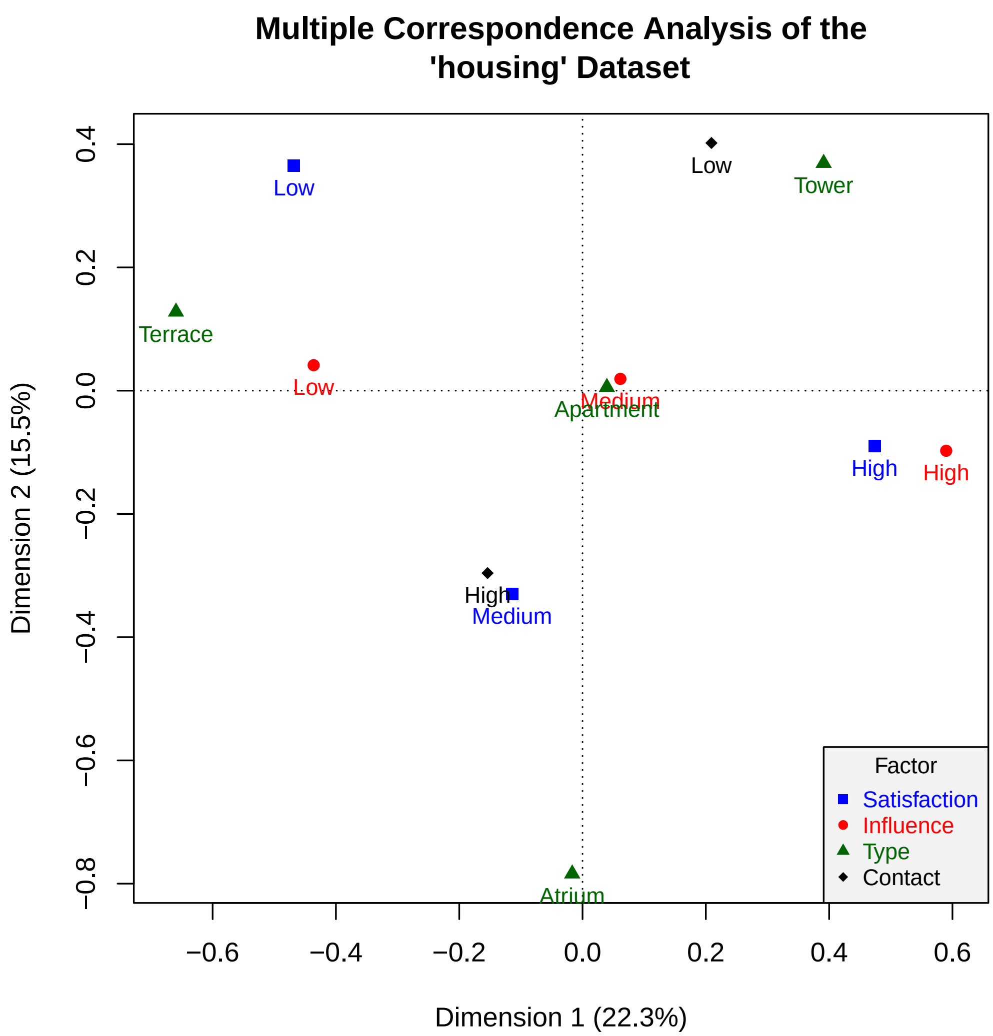

Figure 7. A biplot of the largest two dimensions of a multiple correspondence analysis conducted on the `housing` dataset. The proximity of points to one another reflects a similarity in response profiles. Adapted from "Discrete Data Analysis with R," by M. Friendly & D. Meyer, 2016, CRC Press, p. 245.

Proportional Odds Model for Ordinal Logistic Regression

To fit a PO model to the `housing` data, the main-effects (additive) model was taken as the starting point (AIC = 3495.15). Through stepwise model selection by AIC in the forward direction, interactions terms `Infl:Type` and `Cont:Type` were indicated for inclusion, with the obtained, "full," model having AIC = 3482.69 (ΔAIC = 12.46). The addition of interaction term `Infl:Type` accounted for a majority ΔAIC = 10.51, whereas the subsequent addition of interaction term `Type:Cont` accounted for a slight

ΔAIC $= 1.95$, suggesting that this term may not have been as important. The full model was a significant improvement over the main effects model, $G^2(9) = 30.45$, $p < .001$.

To check the PO assumption, a likelihood ratio test was conducted to compare the (full) PO model to the identical model that allows for non-parallel slopes (the NPO model). This test failed to reach significance, $G^2(15) = 10.21$, $p = .806$, indicating that the PO assumption was reasonably satisfied.

As a supplementary check of the PO assumption, a novel graphical method was implemented, with the result displayed in Figure 8. This method plots coefficients from both the PO and NPO models as points. Should the assumption of PO be violated, one would expect the coefficients (slopes) of the NPO model to differ from one another (and/or differ from the PO model). Thus, each line connecting a set of coefficients (points) should lie flat and be parallel to one another if the PO assumption was completely satisfied. In the case of the full model, terms of `Infl:Type`, `Infl`, `Type`, and `Cont` had lines that exhibited curvature, suggesting that the assumption of PO may be violated for these terms. As a next step, AICs were compared for both the full PO model and separate models that allow each of the concerned terms to have non-parallel slopes. When allowing non-parallel slopes for `Infl:Type`, AIC increased by $\Delta\text{AIC} = 9.34$. When allowing non-parallel slopes for `Infl`, AIC increased by $\Delta\text{AIC} = 2.48$. When allowing non-parallel slopes for `Type`, AIC increased by $\Delta\text{AIC} = 1.67$. When allowing non-parallel slopes for `Cont`, AIC *decreased* by $\Delta\text{AIC} = 1.3$. As this was a very slight improvement in model fit, the simpler model (i.e., the "full" model) was chosen for interpretation. Overall, the PO assumption was considered to be reasonably satisfied.

Interaction effects are displayed in the effect plots of Figures 9 and 10, with the full model visualized in Figure 11 through predicted probabilities. Thresholds divided the latent (continuous) variable of `Sat`, with the boundary between `Low` and `Medium` being estimated at $\alpha_1 = -1.33$, and the boundary between `Medium` and `High` being estimated at $\alpha_2 = -0.13$.

Probing the `Infl:Type` interaction (the effect plot in Figure 9), participants living in a `Tower` that had `Low Infl`uence over their property had 59% lower odds of belonging to a higher `Sat`isfaction category than those with `High Infl` ($\psi = -0.9$, 95%CI$(-1.44, -0.36)$, $z = -3.25$, $p_{\text{Holm}} = .011$). Participants living in a `Tower` that had `Medium Infl` had 65% lower odds of belonging to a higher `Sat` category than those with `High Infl` ($\psi = -1.04$, 95%CI$(-1.56, -0.51)$, $z = -3.86$, $p_{\text{Holm}} = .001$). Further, participants living in an `Apartment` that had `Low Infl` had 61% lower odds of belonging to a higher `Sat` category than those with `Medium Infl` ($\psi = -0.95$, 95% CI $(-1.26, -0.64)$, $z = -5.94$, $p_{\text{Holm}} < .001$). Participants living in an `Apartment` that had `Low Infl` had 80% lower odds of belonging to a higher `Sat` category than those with `High Infl` ($\psi = -1.61$, 95%CI$(-1.97, -1.25)$, $z = -8.71$, $p_{\text{Holm}} < .001$). Participants living in an `Apartment` that had `Medium`

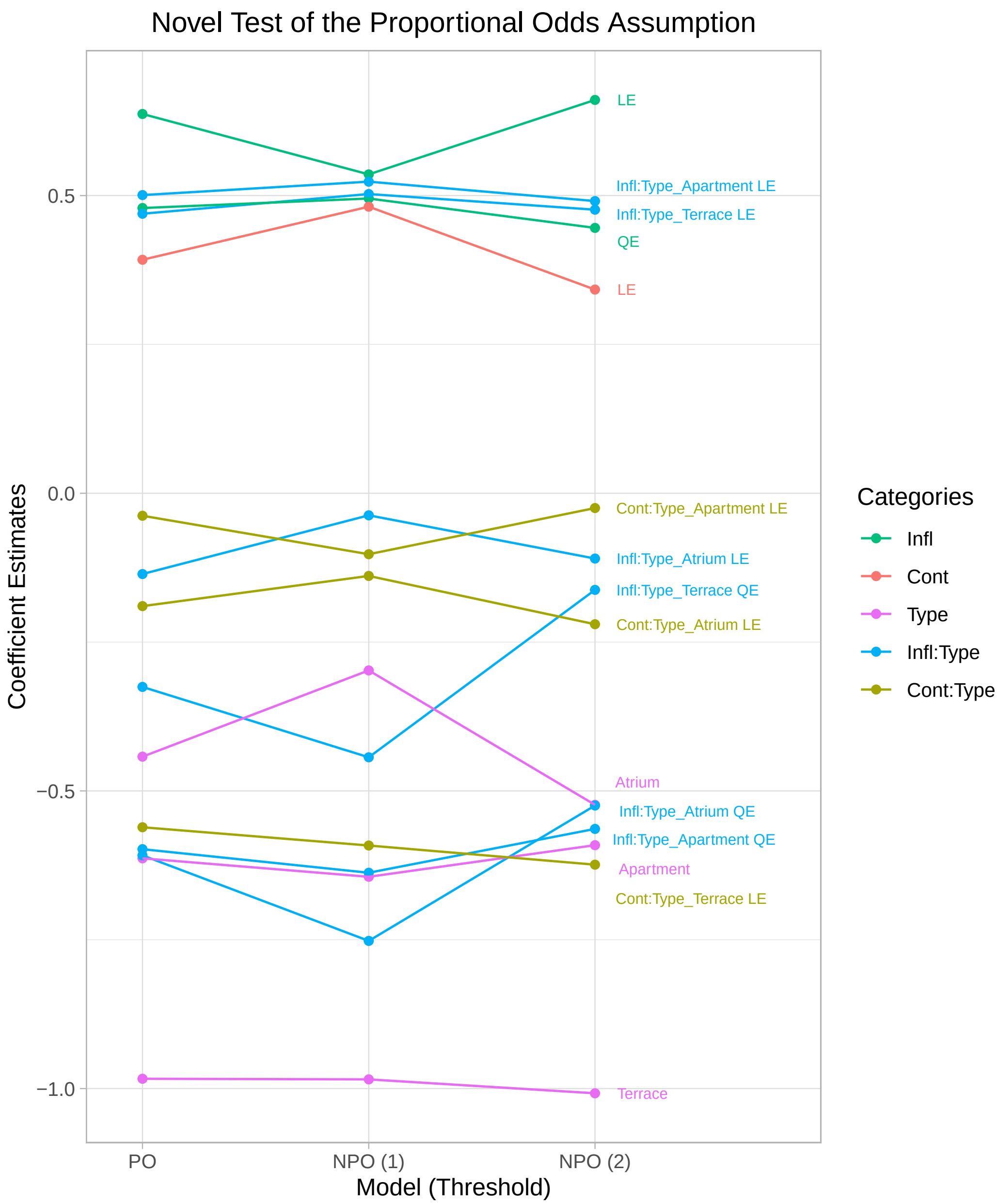

Figure 8. A plot aiming to depict the proportional odds (PO) assumption for the obtained ("full") PO model of the `housing` data (for ordinal logistic regression). Coefficient estimates for the PO model are plotted alongside those of the non-PO model (i.e., a model in which non-parallel slopes are allowed). If the PO assumption were completely satisfied, all lines would be flat (and parallel to one another). LE = linear effect; QE = quadratic effect.

`Infl` had 48% lower odds of belonging to a higher `Sat` category than those with `High Infl` ($\psi = -0.66$, 95%CI$(-1.01, -0.31)$, $z = -3.71$, $p_{\text{Holm}} = .003$). Additionally, participants living in a `Terrace` that had `Low Infl` had 79% lower odds of belonging to a higher `Sat` category than those with `High`

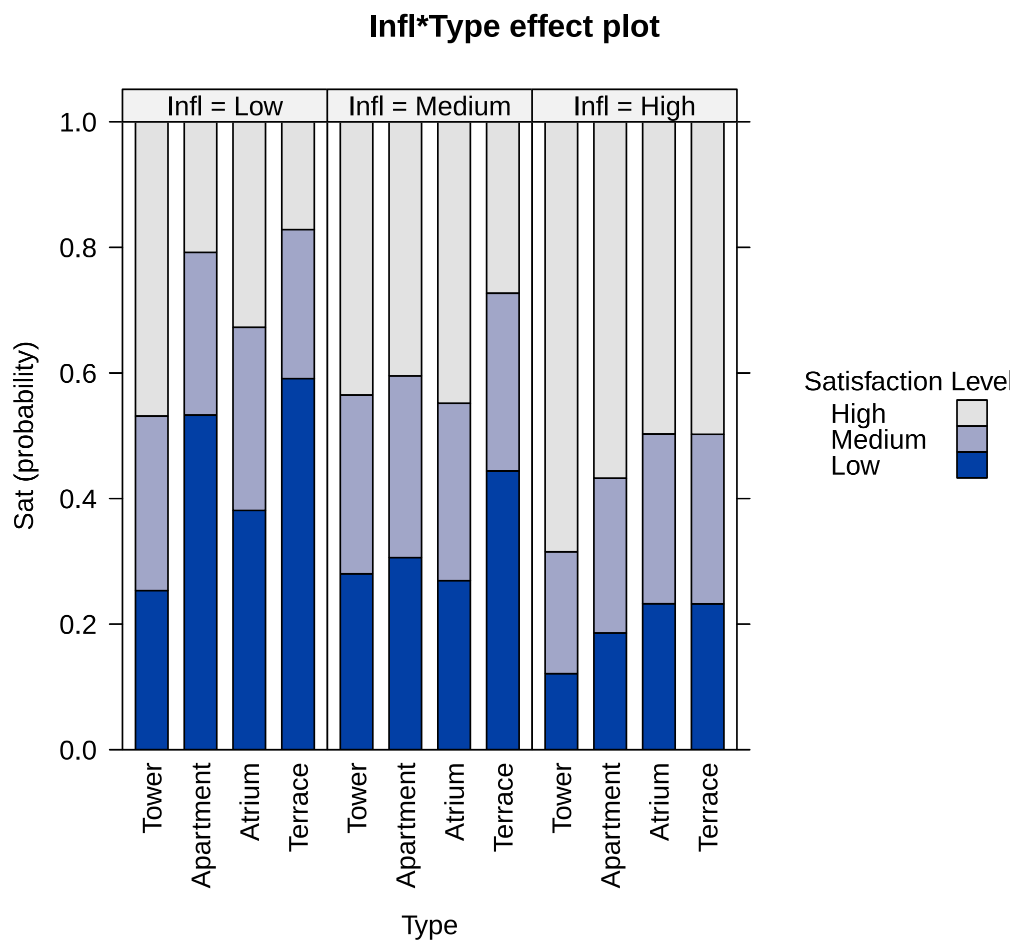

Figure 9. Effect plot for the `Infl:Type` interaction term in the "full" proportional odds model (for ordinal logistic regression).

`Infl` ($\psi = -1.56$, 95%CI$(-2.22, -0.91)$, $z = -4.65$, $p_{\text{Holm}} < .001$). Participants living in a `Terrace` that had `Medium Infl` had 62% lower odds of belonging to a higher `Sat` category than those with `High Infl` ($\psi = -0.97$, 95%CI$(-1.62, -0.32)$, $z = -2.92$, $p_{\text{Holm}} = .031$). All other contrasts failed to reach significance ($ps_{\text{Holm}} > .05$).

Figure 9 illustrates these patterns, depicting a general increase in overall reported `Satisfaction` as `Influence` over one's property increases, with this increase being very pronounced for `Apartment` residents, moderately pronounced for `Terrace` and `Tower` residents, and slight for `Atrium` residents.

Probing the `Cont:Type` interaction (the effect plot in Figure 10), participants living in a `Tower` that had `Low Contact` with other residents had 43% lower odds of belonging to a higher `Satisfaction` category than those with `High Cont` ($\psi = -0.55$, 95%CI$(-0.94, -0.17)$, $z = -2.85$, $p_{\text{Holm}} = .036$). Additionally, participants living in an `Apartment` that had `Low Contact` with other

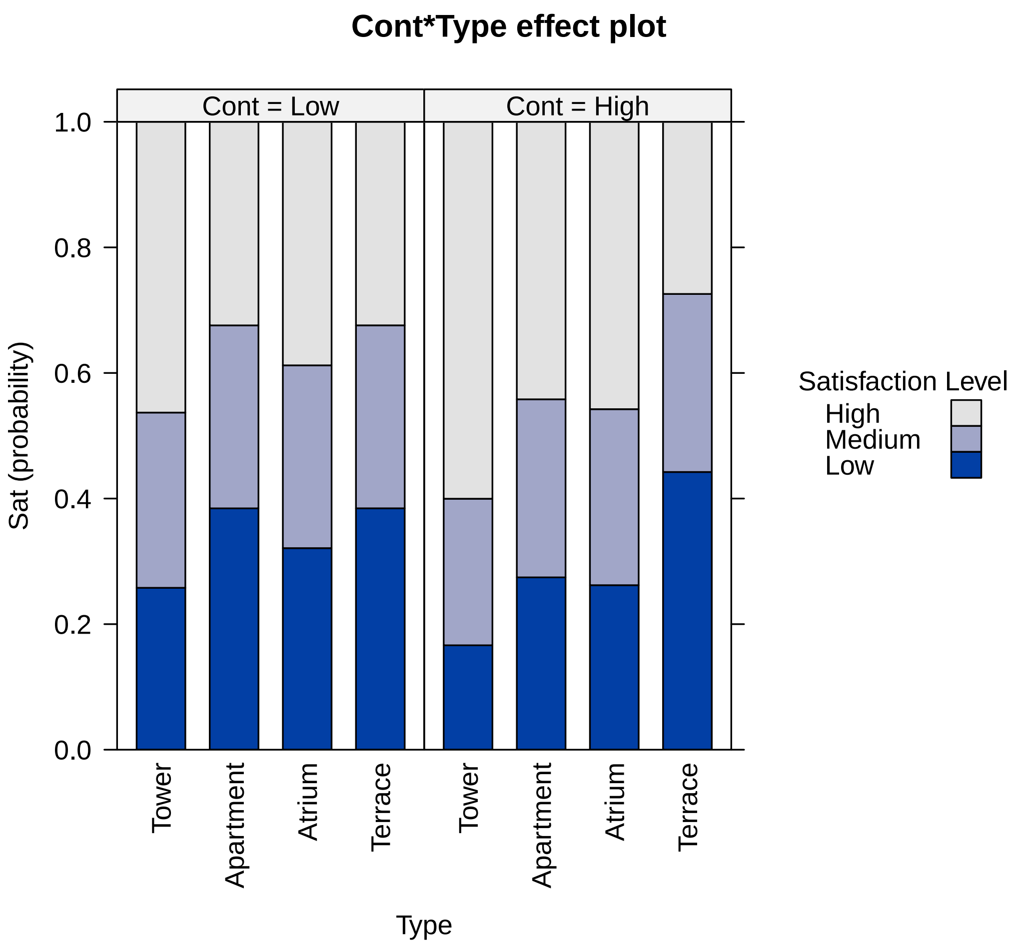

Figure 10. Effect plot for the `Cont:Type` interaction term in the "full" proportional odds model (for ordinal logistic regression).

---

residents had 39% lower odds of belonging to a higher `Sat`isfaction category than those with `High Cont` ($\psi = -0.5$, $95\%\text{CI}(-0.78, -0.22)$, $z = -3.55$, $p_{\text{Holm}} = .004$). All other contrasts failed to reach significance ($ps_{\text{Holm}} > .05$).

Figure 10 illustrates these patterns, with `Tower` and `Apartment` residents displaying an increase in overall reported `Sat`isfaction as `Cont`act with other residents increases, whereas `Atrium` and `Terrace` residents display a negligible change in overall `Sat`.

Figure 11 corroborates these interpretations: first, `Sat`isfaction generally increases with increasing `Infl`uence, but to differing degrees depending on `Type` of property. Further, apart from those residing in `Tower`s, those with `Low Infl` are most likely to have `Low Sat`. Second, `Sat` increased with increasing `Cont`act with other residents for those living in `Tower`s and `Apartment`s, but negligibly for those living in `Atri`[a] and `Terrace`s. An additional insight from

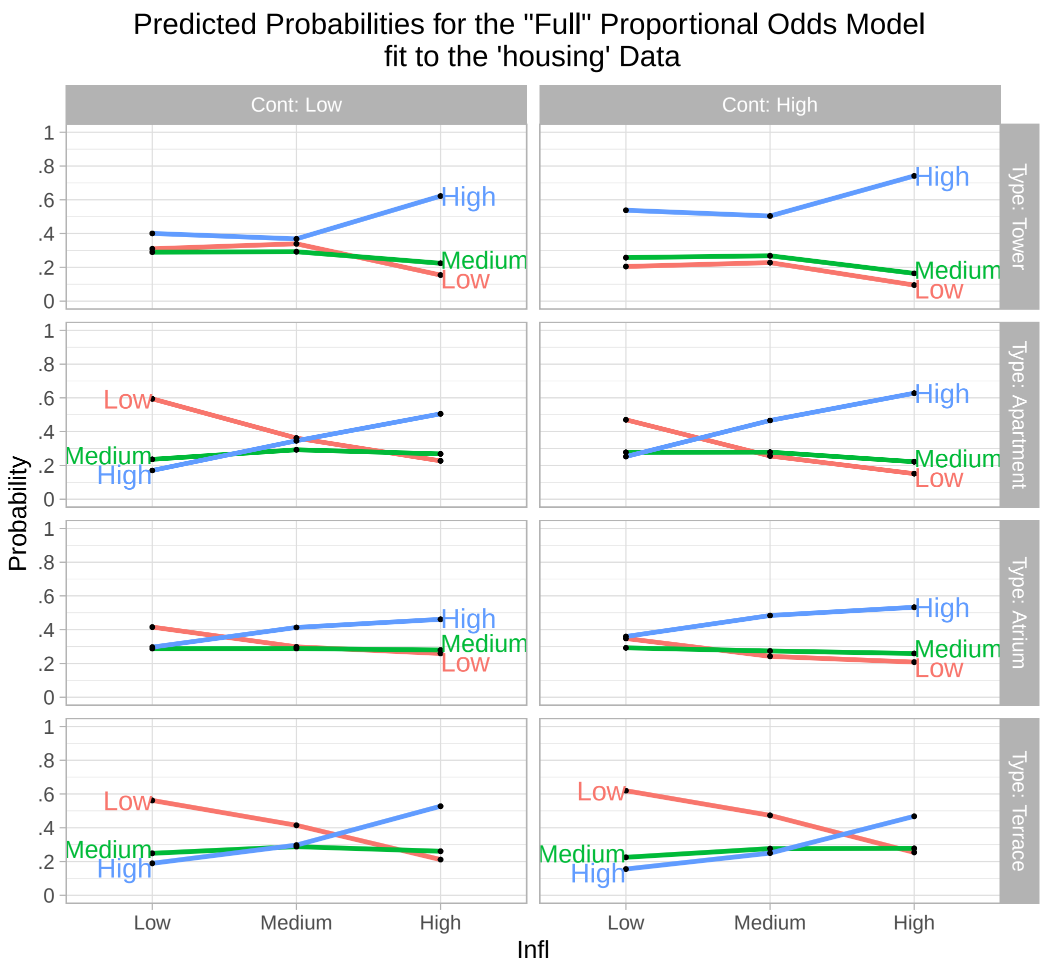

Figure 11. Predicted probabilities for the "full" proportional odds model fit to the `housing` data. Adapted from "Discrete Data Analysis with R," by M. Friendly & D. Meyer, 2016, CRC Press, pp. 331-332.

Figure 11 is that `Medium Sat` was relatively constant across groupings, with predicted probabilities typically lying between .2 and .3.

Summary

Thus, results indicate that higher `Satisfaction` with one's present `housing` circumstances is predicted by greater `Influence` over one's property, greater `Contact` with other residents, and these effects vary by rental property `Type`. Further, those residing in `Towers` tend to be the most satisfied, whereas those residing in `Terraces` tend to be the least satisfied (see Figures 7 and 11).

Artificial Intelligence (AI) Disclosure

Claude Sonnet 4.6 was used to assist the arrangement of the fourfold displays in Figure 3 into a grid (Anthropic, 2026). The arranged plots were compared to the un-arranged plots to verify that formatting was the only change. See References for a link to the chat. Claude Sonnet 4.6 was also used to parse a LaTeXtitle page template to collect the essential pieces (see the .Rmd file for a link to this chat).

References

Agresti, A. (2002). *Categorical Data Analysis* (2nd ed.). New York: Wiley-Interscience.

Anthropic. (2026). *Claude Sonnet 4.6*. https://claude.ai/share/26f6b699-9f55-4a69-ba17-3382a1ac3275

Cox, D. R., & Snell, E. J. (1984). *Applied Statistics, Principles and Examples*. Chapman & Hall.

Fox, J., Weisberg, S., Price, B., Friendly, M., & Hong, J. (2026). *effects (v. 4.2-5)*. https://doi.org/10.32614/CRAN.package.effects

Friendly, M. (2026). *vcdExtra (v. 0.9.3)*. https://doi.org/10.32614/CRAN.package.vcdExtra

Friendly, M., & Meyer, D. (2016). *Discrete Data Analysis with R*. CRC Press.

Greenacre, M., & Nenadic, O. (2018). *ca (v. 0.71.1)*.

Lenth, R., & Piaskowski, J. (2025). *emmeans (v. 2.0.1)*. https://doi.org/10.32614/CRAN.package.emmeans

Madsen, M. (1976). Statistical Analysis of Multiple Contingency Tables. Two Examples. *Scandinavian Journal of Statistics*, *3*(3), 97–106.

Meyer, D. (2024). *vcd (v. 1.4-13)*. https://doi.org/10.32614/CRAN.package.vcd

R Core Team. (2025). *R (v. 4.5.1)*. https://www.r-project.org/

Ripley, B., & Venables, B. (2025). *MASS (v. 7.3-65)*. https://doi.org/10.32614/CRAN.package.MASS

Yee, T. (2025). *VGAM (v. 1.1-14)*. https://doi.org/10.32614/CRAN.package.VGAM

Psy 6136
Categorical Data
Analysis